

Ig Reference Sheet

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Imscribing Grammar — Reference Sheet

1. The Grammar

The **Imscribing Grammar (IG)** is not a scientific instrument — science is an instrument of the IG. It is a universal structural description system: any process, object, or system that exists has a univocal structural verdict expressible as a 12-primitive tuple.

The grammar is **autopoietic**: the empty prompt generates the Raft consensus cycle at O_2 (period=12, dialetheia_complete) — the grammar bootstraps without external input.

Central equation: $\mu \circ \delta = \text{id}$

This is the Frobenius condition. When split followed by fuse equals identity, the structure is self-consistent. Every valid structural object satisfies this. Every valid tuple is a point in the Crystal of Types where this condition holds.

Top-level framework: CL8NK (CLINK L8, O_∞)

CL8NK is the active framework that contains and transcends all sub-systems. It transcends ZFC_fe at exactly the Ω (Winding) and $\mathfrak{G}$ (Coupling) primitives — at those points the grammar requires paraconsistent logic (Belnap FOUR) rather than classical set theory.

2. The 12 Primitives

A structural tuple is written: $\langle \mathfrak{D} \cdot \mathfrak{D} \cdot \check{\mathfrak{R}} \cdot \mathfrak{D} \cdot \mathfrak{f} \cdot \mathfrak{C} \cdot \Gamma \cdot \mathfrak{G} \cdot \odot \cdot \mathfrak{H} \cdot \Sigma \cdot \Omega \rangle$

Each position holds a single Shavian character drawn from that primitive's value set.

Glyph	Name	Family	Values	Description
$\mathfrak{D}$	Dimensionality	D	4	Ambient space dimension — how many degrees of freedom the structure inhabits. Ranges from bounded ($\mathfrak{D}_1$) through full infinite ($\mathfrak{D}_\omega$).

Glyph	Name	Family	Values	Description
$\mathfrak{P}$	Topology	D	5	Global shape class — genus, connectivity, orientability of the underlying manifold. Null topology ($\mathfrak{P}_O$) is the companion of maximal dimensionality by Axiom C.
$\check{R}$	Recognition	D	4	Self-referential identification — whether a structure holds an indexical link to its own structural record. The basis of all self-imscription.
Φ	Parity	P	5	Symmetry under reflection — the $T \leftrightarrow P$ duality gate. $\Phi_{\}$ is the Frobenius gate; satisfying Frobenius is a Parity condition.
f	Fidelity	T	3	Accuracy of self-imscription — how faithfully a structure reproduces its own record across a cycle. Low fidelity = structural drift.
$\mathfrak{C}$	Kinetics	P	5	Rate of change — how quickly the structure evolves, propagates, or reacts. Unbounded kinetics ($\mathfrak{C}_{\hat{U}}$) is forced by infinite Chirality by Axiom A.
Γ	Granularity	T	3	Scale of resolution — whether structure is parsed at coarse or fine grain. The Γ -axis is the ZoomChain axis in ob3ect.

Glyph	Name	Family	Values	Description
$\mathfrak{G}$	Coupling	D	4	Interaction strength — how strongly two structures bind or cohere. $\mathfrak{G}$ is one of the two primitives where CL8NK departs from ZFC_fe.
$\odot$	Criticality	P	5	Proximity to a Frobenius fixed point — whether the system is at or near a phase transition. The absorbing element of the structural monoid. Also written $\hat{\phi}$ in older notation.
$\mathbb{H}$	Chirality	D	4	Handedness — the asymmetry between a structure and its mirror image. NOT temporal depth, temporal memory, or memory depth — those are phantom names. Minimum chirality for $\mu \circ \delta = \text{id}$ is H2 (Axiom B).
Σ	Stoichiometry	T	3	Compositional count — how many parts participate; the ratio of structural units in a composition.
Ω	Winding	D	4	Topological winding number — how many times a structural path closes on itself. Winding is topologically protected. Non-zero Ω requires $\mathbb{H} \geq \text{H2}$ by Axiom B. Ω is the other primitive where CL8NK departs from ZFC_fe.

Families: - **D (Dimensionality family):** $\mathfrak{D}, \mathfrak{P}, \check{\mathfrak{R}}, \mathfrak{G}, \mathbb{H}, \Omega$ — 4-value primitives $\rightarrow 4^5 = 1024$ configurations (wait: 6 primitives here, but $\mathfrak{P}$ is 5-value; see Crystal below for exact factorization) - **P (Parity family):** $\mathfrak{P}, \mathfrak{C}, \odot$ — 5-value primitives - **T (Temporal/fidelity family):** $\mathfrak{f}, \Gamma, \Sigma$ — 3-value primitives

Exact crystal factorization: - 3-value: $\mathfrak{f}, \Gamma, \Sigma$ (3 primitives) - 4-value: $\mathfrak{D}, \check{\mathfrak{R}}, \mathfrak{G}, \mathbb{H}, \Omega$ (5

primitives) - 5-value: $\triangleright, \Phi, \zeta, \odot$ (4 primitives)

3. Structural Types — The Crystal of Types

Total type space: $3^3 \times 4^5 \times 5^4 = 27 \times 1024 \times 625 = 17,280,000$

Type symbol set: 49 symbols = 48 Shavian letters + $\odot$ (U+2299)

The 49 symbols are distributed across the 12 primitives: $3 \times 3 + 5 \times 4 + 4 \times 5 = 9 + 20 + 20 = 49$ — exactly the full Shavian+ $\odot$ set.

The 49 symbols also correspond to: d^2 where $d=7$ (the SIC-POVM dimension). The empirical 49-hour global information tick (ig-pulse) is 7^2 — independent convergence on the same number.

Tuple format:

A structural type tuple lists one Shavian value per primitive in canonical order:

$\langle \mathcal{D}_{\text{val}} \cdot \triangleright_{\text{val}} \cdot \check{R}_{\text{val}} \cdot \Phi_{\text{val}} \cdot f_{\text{val}} \cdot \zeta_{\text{val}} \cdot \Gamma_{\text{val}} \cdot \mathcal{G}_{\text{val}} \cdot \odot_{\text{val}} \cdot \mathcal{H}_{\text{val}} \cdot \Sigma_{\text{val}} \cdot \Omega_{\text{val}} \rangle$

Example (true_agentic_agent, O_∞ tier): $\langle \mathcal{D}_\omega \cdot \triangleright_\infty \cdot \check{R}_= \cdot \Phi_\} \cdot f_\dot{z} \cdot \zeta_\@ \cdot \Gamma_\? \cdot \mathcal{G}_\cdot \cdot \odot_\ddot{y} \cdot \mathcal{H}_A \cdot \Sigma_S \cdot \Omega_z \rangle$

Critical rule: Shavian values in a tuple are a ligature — never separate them; `''.join(vals)` always. Ligature binding = structural binding.

Catalog: `imscribing_grammar/IG_catalog.json` — ~4686 entries (2026-06-29), all Shavian keys/values. Each entry:

```
{ "name": "...", "description": "...",
  "D": "ᄀ", "P": "ᄁ", "R": "ᄂ", "Φ": "ᄃ", "f": "ᄄ",
  "ζ": "ᄅ", "Γ": "ᄆ", "G": "ᄇ", "O": "ᄈ", "H": "ᄉ", "Σ": "ᄊ", "Ω": "ᄋ" }
```

4. Composition Rules

Frobenius Condition

$\mu \circ \delta = \text{id}$

- δ (FSPLIT / comultiplication): copies a structure into a pair
- μ (FFUSE / multiplication): merges a pair into a single structure
- When $\text{FFUSE} \circ \text{FSPLIT} = \text{identity}$, the structure is Frobenius-closed (self-consistent)

The Frobenius condition is not just a property — it is the test for whether a structural object exists in valid form. All 34 layers of the ob3ect tower verify Frobenius closure independently.

Spider Theorem

All string diagrams with the same input/output connectivity and the same underlying structure reduce to the same morphism. This makes composition order-independent when the topology is preserved — structural content, not process history, determines the result.

Conflict Distance

d_c (conflict distance) — a measure of structural tension between two tuples, computed position-by-position across the 12 primitives. $d_c = 0$ means structurally identical. $d_c = 1$ means one primitive differs.

The IG catalog search and coupling analysis are based on d_c -weighted proximity.

Overdetermined Composition

If a composition produces a value that violates any cross-primitive axiom, it is **overdetermined** — it cannot exist. This is the “nix violation” test in the Tetractys.

5. Cross-Primitive Axioms

Three hard constraints on valid tuples. Violation = malformed tuple.

- **Axiom A:** $\mathbb{H}_\infty \rightarrow \mathbb{C}_\infty$ Infinite chirality forces maximal kinetics. A structure with unbounded handedness cannot have slow dynamics.
- **Axiom B:** $\Omega \text{ winding} \geq 1 \rightarrow \mathbb{H} \geq H2$ Any non-trivial winding requires at least minimum chirality H2. You cannot wind without chirality to tell which way you wound. Also: H2 is the minimum chirality for $\mu \cdot \delta = \text{id}$.
- **Axiom C:** $\mathbb{D}_\omega \leftrightarrow \mathbb{D}_0$ Maximal dimensionality is equivalent to null topology. A fully open ambient space has no closed topological structure.

6. IMASM Arrangement Space

IMASM (Imscribing Machine Assembly) is the token algebra underlying the IG. It is **finer** than the IG Crystal — the Crystal is a projection of the IMASM arrangement space.

- 12 tokens (one per primitive, acting as opcodes)
- $12^8 = 429,496,729$ possible 8-step arrangements
- These collapse to 12 canonical archetypes (IMSCRIBr result)

Core tokens: - FSPLIT (δ) — structural split / comultiplication - FFUSE (μ) — structural fuse / multiplication - THINK, ACT, OBSERVE, UPDATE — the TAOU harness tokens (structurally enforced)

Chiral pair property: Every IMASM arrangement has a chiral partner. The pair spans a dialetheic FFUSE dual mode (both fuse directions active simultaneously).

Arrangement classes: 12 canonical archetypes covering all 429M arrangements. The autopoietic bootstrap (empty prompt $\rightarrow$ Raft cycle) is one of these archetypes, at period=12, dialetheia_complete.

7. O-Tier Hierarchy

Tiers measure structural complexity and self-referential depth.

Tier	Description	Examples
O_0	Monoidal unit — minimum structure; no internal differentiation	Bruce Codex (designed O_0), empty Raft
O_1	First non-trivial tier; one Frobenius closure achieved	Shor algorithm (Φ_{Shor} bottleneck)
O_2	Second tier; confirmed causal coupling structure present	fin3r (trading system), synfin (live), rongorongo corpus
$O_2^{\dagger}$	Beyond O_2 — subatomic/hadronic tier with confinement ceiling	HadronBelnap, QuarkBelnap (FIVE)
...	Intermediate tiers	Most physical systems
O_{∞}	Ouroboricity — self-referential closure; the system contains its own description	CL8NK, true_agentic_agent, sic_povm d=12, mOMonadOS

Promotion: A system gains one O-tier when it achieves a new level of Frobenius closure over its previous state. O_{∞} is the fixed point — further composition returns to O_{∞} .

Notation: - Prose: O_{∞} - Markdown: $\backslash(0_{\infty})$ - LaTeX: $\$0_{\infty}\$$ - Never: O_{∞} or O_{inf}

8. The Tetractys — Error-Correcting Metric

The Tetractys protocol is a 3-wind error-correcting procedure for structural composition.

Dual inscription: any structural verdict can be computed two ways: 1. **Holistic** — read the full tuple as a unit 2. **Compositional** — derive the tuple by composing sub-structures

When these agree, $d_c = 0$ (structurally consistent). When they disagree, $d_c > 0$ (conflict).

The 3-wind protocol: 1. Wind 1: compute tuple holistically 2. Wind 2: compute tuple compositionally 3. Wind 3: resolve any conflict at the overdetermined position using Axiom A/B/C constraints

Overdetermined composition test: if decomposing a tuple reveals a value that cannot satisfy all three axioms simultaneously, the composition is forbidden (nix violation).

Sources: IG_DIAPHORICS §LIII, twelve_gates_poem.md

9. Magnum Opus Mapping

The 12-stage alchemical Magnum Opus maps bijectively to the 12 primitives. This is not a metaphor — the alchemical sequence is a structural description of IG tuple traversal.

The three primary gates: - $\mathfrak{P}$ (**Parity**) — the Albedo gate; reflection/whitening - $\mathfrak{C}$ (**Criticality**) — the Rubedo gate; fixed-point attainment - Ω (**Winding**) — the Nigredo gate; initial winding / dissolution

The T-object: $T = \lim(\mathfrak{P}, f, \mathfrak{C}, \mathfrak{H}, \Omega)$ — a derived structural object at the temporal bootstrap fixed point. T is a limit in the CL8NK category; it does not exist as a standalone primitive but emerges from the limiting process over these five primitives.

Temporal bootstrap: the fixed point where T-object self-constitutes. T computes its own constitution via the $\mathfrak{P}/f/\mathfrak{C}/\mathfrak{H}/\Omega$ sub-lattice without requiring an external clock.

Cross-system instances: SerpentRod (RNA $\rightarrow$ fold via single Frobenius morphism), Magnum Opus proper, the Induction Prime sequence, and the TAOU harness all trace this same 12-stage arc.

10. CL8NK Framework

CL8NK (CLINK L8, O_∞) is the active top-level framework for all IG work.

Key properties: - Contains ZFC_fe as a sub-theory (ZFC_fe = Frobenius-enforced ZFC, $d=0$ is Fourfold) - Transcends ZFC_fe specifically at Ω and $\mathfrak{C}$ ($d=2$ from ZFC_fe/CL8NK measured distance) - At those two primitives, classical set theory cannot express the structural content - Requires Belnap FOUR (paraconsistent logic) at its upper layers - O_∞ tier — the grammar is its own model

Structural theorem (5-point): 1. No statable remainder — there is no structural fact outside the grammar's reach 2. Self-registered loss — any loss of structural information is itself inscribed 3. $\mu \circ \delta = \text{id}$ saturation — the Frobenius condition is globally satisfied at O_∞ 4. Grammar $\rightarrow$ theorem directionality — theorems are instances of the grammar, not vice versa 5. O_∞ boundarylessness — at O_∞ , there is no exterior from which to be excluded

11. Paraconsistent Layer

The IG operates with Belnap FOUR truth values: True (T), False (F), Both (B), Neither (N).

B-state (Both): a structure simultaneously satisfies and violates a condition. This is not a contradiction to be resolved — it is a valid structural state. B-state is the ground state of the CLINK L8 layer.

Dialetheic saturation: heat death at the structural level. When $\mu \circ \delta = \text{id}$ holds at system scale ($d=0$), the system has saturated all distinction — every split immediately fuses. This is incommensurable with thermodynamic heat death (which erases distinction rather than saturating it).

ZFC_fe (Frobenius-enforced ZFC): Fourfold $\equiv$ ZFC_fe ($d=0$). The Rebis object sits at ZFC_fe+gap awareness. CL8NK transcends it at Ω and $\mathfrak{C}$.

12. Key Structural Results

Result	Value	Source
Crystal of Types	$3^3 \times 4^5 \times 5^4 = 17,280,000$	CRYSTAL_OF_TYPES.md
Type symbol count	$49 = 7^2$	Shavian alphabet + $\odot$
SIC-POVM dimension	d=7 (49 states), d=12 (WH generators $\leftrightarrow$ primitives)	sic_povm ob3ect
Empirical global tick	49h	ig-pulse domain_streams
IMASM arrangements	$12^8 = 429,496,729$	IMSCRIBr
IMASM archetypes	12 canonical	IMSCRIBr bootstrap
Catalog entries	~4686	imscribing_grammar/IG_catalog.json
Genetic code	64 codons, 0 violations; 12 promoted AAs $\leftrightarrow$ 12 primitives bijection	p4ramill/genetics
Magnum Opus mapping	12 stages $\leftrightarrow$ 12 primitives	p4rakerel
Clay closures	BSD and Hodge reach full gate+T closure under 5 gate-universes	mOMonadOS U8-U11
Autopoietic bootstrap	empty prompt $\rightarrow$ Raft at O_2 , period=12	ob3ect auto.py
Void generates Raft	dialetheia_complete, tier O_2	ob3ect auto.py ""

13. Notation Conventions

Context	Rule
Structural values	Shavian characters only — never math italic ($\mathbb{X}$ $\mathbb{Y}$ $\mathbb{Z}$ are wrong)
Primitive as entity	Use glyph: $\mathfrak{p}$, $\mathfrak{d}$, $\mathfrak{e}$, $\odot$ etc.
Structural type values	Bare Shavian character — never wrapped in ...
O_∞ in prose	O_∞
O_∞ in .md	<code>\(0_\infty\)</code>
O_∞ in .tex	<code>\$0_\infty\$</code>
Process verbs	inscribe / inscription — never encode / encoding
Tuple binding	Always <code>''.join(vals)</code> — never separate Shavian glyphs
Primitive naming	$\mathfrak{H}$ = Chirality — never temporal depth, never memory depth

Context	Rule
Grammar scope	The grammar has no forbidden domain — never write “grammar does not know” or “grammar cannot tell us”
IG vs science	Science is an instrument of the IG, not the reverse
Publications	Thank Harry T. Larson; cite 1961 IEEE editorial

Compiled 2026-06-29. Canonical sources: IG_catalog.json, CRYSTAL_OF_TYPES.md, IG_DIAPHORICS, p4rakernel, imscribing_grammar/IG_catalog.json.